Taksh Kalathia

+1 (551) 330-1040 | tdkalathia55@gmail.com | linkedin.com/in/taksh05 | github.com/takshkalathia

EDUCATION

Northeastern University, Khoury College of Computer Science, Boston, MA, USA **Sept 2026 – Dec 2028**
Master of Science in Computer Science

Nirma University, Ahmedabad, Gujarat, India **Sept 2022 – May 2026**
Bachelor of Technology in Electrical Engineering with Minor in Data Science

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java, HTML.
- **AI/ML:** Metaheuristic Optimization, PyTorch, TensorFlow, Keras, scikit-learn.
- **Cloud & Data Science:** Tableau, Power BI, NumPy, pandas, Matplotlib, Google Colab, AWS, GCP, MongoDB, SQL.
- **Engineering:** MATLAB, Statistics, Mathematics, Digital Electronics, IoT, Renewable Energy, AutoCAD, RPi, ESP32.

EXPERIENCE

Research Intern | Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat **Jan 2026 – Apr 2026**

- Developed an **AI-based optimization framework** for a Solar–Wind–Battery Hybrid Renewable Energy System (HRES), using Python, machine learning, and metaheuristic optimization techniques to select cost-effective system configurations.
- Modeled **solar PV, wind generation, battery State of Charge, energy dispatch, and system costs** using 3,546 hourly real-world renewable energy data points to evaluate feasible microgrid configurations and minimize **Net Present Cost**.
- Implemented and benchmarked **12 metaheuristic optimization algorithms**, including PSO, GWO, ABSO, GA, and IPSO, using Python and benchmark functions to compare convergence behavior, solution quality, and optimization performance.
- Identified **PSO-based optimization** as an effective approach for HRES sizing, obtaining an optimized configuration of **303 PV units, 257 wind turbines, and a 2,870 kWh battery** with an estimated **NPC of $1.95M**, providing a cost-oriented solution for renewable microgrid planning and supporting **two conference paper submissions**.

Intern - Supplier Quality Engineer, E&E Dept. | Mahindra & Mahindra Ltd., Pune **Mar 2025 – Jun 2025**

- Analyzed supplier quality data for **electrical and electronic components** using Excel, identifying defect patterns, recurring quality issues, and manufacturer-wise trends to support supplier quality evaluation and corrective actions.
- Performed **quality inspection and root-cause analysis** using techniques such as **8D problem solving, Pareto analysis, and defect-trend analysis** to investigate recurring component and process-related quality issues.
- Collaborated with **Tier-1 and Tier-2 suppliers** during supplier evaluations and process audits, investigated quality issues, and supported the development of **corrective and preventive actions (CAPA)** to address identified root causes.
- Developed interactive **Power BI dashboards** to visualize monthly quality metrics, defect trends, and supplier performance, enabling data-driven identification of critical quality issues and supporting SQE decision-making.

PROJECTS

Lung Cancer Detection Using AI | Python, AI/ML, CNN, MobNetV1/V2, EfficientNetB0

- Developed a **TinyML-assisted lung cancer detection framework** using CT scan images to enable accurate and resource-efficient **Benign vs. Malignant** classification for potential edge deployment.
- Trained and evaluated **MobileNet-V1, MobileNet-V2, EfficientNetB0**, and a **CNN with MobileNet-V3 transfer learning**, comparing **RMSprop, Adam, and Nadam** using accuracy, precision, recall, ROC-AUC, and confusion matrix analysis.
- Identified **MobileNet-V2 with RMSprop** as the best-performing configuration, achieving **97.78% accuracy** before quantization and **97.92% after quantization**, with a ROC-AUC of **0.997**.
- Applied **post-training quantization** to reduce the MobileNet-V2 model size from **18.89 MB to 2.39 MB (87.35% reduction)**, addressing memory constraints for **TinyML and resource-constrained edge deployment**.

COVID-19 India Data Analytics Dashboard | Tableau, Data Analysis, Data Visualization

- Developed an interactive **Tableau dashboard** to analyze COVID-19 trends across Indian states using datasets covering **confirmed cases, recoveries, deaths, testing, and vaccination**, with state-wise, temporal, demographic, and geographic visualizations.
- Applied **calculated fields, interactive filters, geographic mapping, and clustering** to identify regional patterns and relationships, transforming multiple datasets into **actionable data-driven insights** for comparative analysis.

IoT-Based Smart Energy Meter | ESP32, Blynk IoT, EmonLib, IoT

- Developed an **IoT-based smart energy meter** using **ESP32**, **ACS712**, and **ZMPT101B** to measure **RMS voltage, RMS current, apparent power, and energy (kWh)** using EmonLib.
- Integrated **Blynk IoT 2.0** with Wi-Fi for real-time smartphone/web monitoring of electrical parameters and household energy consumption.